



Figure 1: Robot action residuals have input-dependent scale, and one global scale can fail. Residuals are measured against the mean action of eight nearest states from other episodes; binary grippers are excluded, while GR1 hand joints are retained. (a) Tool-Hang chunks grouped by first-half residual RMS retain separated radii over the held-out second half. (b) The held-out Q5/Q1 RMS separation repeats across five datasets (multi-task entries show the median task-wise ratio). (c) Under episode-level cross-fitting, query actions never enter their scale estimates; unseen residual RMS grows monotonically with scale predicted from proprioceptive state (colors follow (b)). (d) With the Student- t tail and architecture fixed, $\sigma(o)$ improves late-five success by 28 points on Tool-Hang and 10 on Transport-MH; Transport-PH is a negative control. Points are seeds and bars are means.

Figure Layout Check

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1 Data analysis

Robot action residuals do not share a single scale. Figure 1 tests this property without grouping examples by the residuals subsequently displayed, and connects the data diagnostic to a controlled policy ablation.